

B.Sc. 1st Semester (Honours) Examination-2022-23

COMPUTER SCIENCE

Course ID : 11511 Course Code : SH/CSC/101/C-1

Course Title :

Programming Fundamentals C (New)

Programming Fundamentals C/C++ (Old)

Time : 1 Hour 15 Minutes

Full Marks : 25

The figures in the right hand margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Unit-I

1. Answer any five questions :

1×5=5

(a) What is pre-processor directive?

(b) Why and when do we use #include directive?

(c) Write differences between while and do-while loop.

(Turn Over)

- (d) Write differences between float and double data type in "C" Language.
- (e) Write a printf statement to output a floating point variable with up to four digits before the decimal point and 2 digits after the decimal point.
- (f) Write a C statement to correctly round a float variable when it is assigned to an int variable. Your answer should not use any library function.
- (g) What is static variable?
- (h) Define union,

Unit-II

2. Answer **any two** questions :

5×2=10

- (a) Write a recursive program in "C" to calculate GCD of two given numbers. Write the differences between ++a and a++. 4+1
- (b) Write a C program to find the transpose of a matrix. What is dynamic memory allocation? 4+1

(c) Write a program in "C" to print the following pattern :

```

1
2   3
4   5   6
7   8   9   10

```

Write differences between "&" and "|" operators.

4+1

(d) Explain the relational operators available in "C".
When a recursive function is called, what happens in

computer's primary memory?

$2\frac{1}{2} \times 5$

Unit-III

3. Answer **any one** question :

10×1=10

(a) Write a C program to add two complex numbers by passing structure to a function. Write a C program to concatenate two strings without using library function. Name the header file for string manipulating library functions. 5+4+1

(b) Write a C program to count characters words and lines in a text file. Explain any two library functions to read from a file. When fopen() is not able to open a file what does it returns? 6+3+1

Bankura Sammilani College

Internal assessment Examination - 2022

Subject - Computer Science

F.M – 5

Course Code – C-1

Time - 30 Minutes

Answer any five questions.

5x1=5

1. What is variable?
2. Why and when do we use #include directive?
3. Write differences between do-while loop and while loop.
4. Write differences between break and continue statement.
5. What is the output of following for loop? Explain.

```
for (l=1; l<10; l++)  
    l+=4;  
printf("%4d", l);
```
6. How and when do we use constants in a C program?
7. What is meant by operator precedence? Explain with example.
8. What is the advantage of including header file in a C program?

B.Sc. 1st Semester (Honours) Examination-2022-23

COMPUTER SCIENCE (PRACTICAL)

Course ID : 11521 Course Code : SH/CSC/101/C-1P

Course Title :

Programming Fundamentals using C (New)

Programming Fundamentals using C/C++ (Old)

Time : 2 Hours

Full Marks : 15

The figures in the right hand margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Problem-10, Viva + LNB-05

1. Answer any one question :

10×1=10

(a) WAP to print the sum of digits of an integer.

(b) WAP to display Fibonacci series using recursion.

(Turn Over)

(c) WAP to reverse a string.

(d) WAP to compute the sum of the first n terms of the following series

$$S = 1 + 1/2 + 1/3 + 1/4 + \dots$$

(e) Write a program that swaps two numbers using pointers.

(f) W.A.P. to copy the contents of one text file to another file, after removing all white spaces.

(g) W.A.P. to compute the factors of a given number.

(h) Write a menu driven program to implement a simple calculator.
